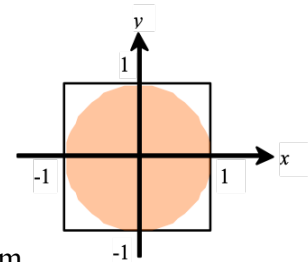


Monte Carlo simulation CS 170 Lab 5

Q1. The Monte Carlo simulation refers to a technique that uses random numbers and probability to solve problems. This method has a wide range of applications in computational mathematics, physics, chemistry, and finance. This section gives an example of using the Monto Carlo simulation for estimation.

Example picture shown on the right.

We can randomly scatter points inside a unit square (side length 1, like above picture). Simply count how many land inside the quarter-circle of radius 1 centered at the origin. Then we can estimate the PI based on the ratio of hits to total points approximates the ratio of areas, like the given form here $PI \approx 4 * \text{numberOfHits} / \text{numberOfTrials}$, since $\text{circleArea} / \text{squareArea} = PI/4$.



Write a program MyPI.java to generate the PI as accurate as possible. Note you can randomly generate n points in square and to see how many points are in circle as well. The larger n is, the more accurate it should be. Your grade won't be judged based on the value of number n, but you can try this program with several different values for n.

Sample output

Tasks (complete in order):

Input/Call	Expected output	Note
n = 100	$PI \approx 3.04$ (rough)	few trials — high variance
n = 10,000	$PI \approx 3.138$	much closer
n = 1,000,000	$PI \approx 3.14159$	very accurate

(10 pts) Identify the geometric condition: for a random point (x, y) in the unit square, write the boolean expression that tells you the point is inside the quarter-circle of radius 1.

(20 pts) Implement the for loop counting body: declare an integer n (your chosen number of trials) and a counter hits. Write a forloop that runs exactly n times — use n as the loop bound. Inside the body, generate a random x and y and increment hits when the circle condition holds.

(10 pts) Compute and print the estimate: apply the formula $\pi \approx 4.0 * \text{hits} / n$. Print your estimate and Math.PI side by side.

(10 pts) Run the program with at least three different values of n (e.g. 100, 10 000, 1 000 000). Record the output in comments at the bottom of your file and describe the trend you observe.